

Araz Mehrabani

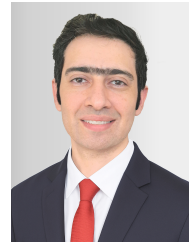
Mechanical Engineer | Design · Product Development · CAD

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📅 Date of birth: 28.07.1989



PROFILE & CORE COMPETENCIES

Mechanical engineer with several years of industrial experience in **machinery design and product development**. Hands-on work from component/assembly design and sizing through tolerancing/manufacturing drawings to shop-floor support, design revisions and prototype optimization.

TECHNICAL FOCUS

Mechanical Design & CAD: CATIA V5, SOLIDWORKS; part/assembly design, machine and drive concepts, detail design, design revisions.

Drawings & Tolerancing: 2D manufacturing drawings, dimensional tolerances, selective GD&T, shaft/bearing fits, bills of materials (BOM).

Mechanical Sizing: material selection, shaft/spring calculations, mechanisms, gear ratio, torque/speed, motor/actuator selection, DFM/manufacturing support.

Prototyping/Additive Manufacturing: STL, Slic3r, G-code, Arduino/Marlin, stepper calibration, ABS/PLA, layer height/infill/temperature/speed.

PROFESSIONAL EXPERIENCE

Development Engineer — Sabaniroo, Tehran

01/2022–02/2023

- Mechanically developed a two-axis **PV-panel cleaning system**: rail/wheel guidance and belt-driven carriage motion, torque/speed/ratio and wheel/rail-load calculations, motor selection, CAD and prototype testing; approx. **10% lower material use**.

Mechanical Engineer — TehranRigzar, Tehran

03/2017–12/2021

- Designed conveyors, jaw crushers, sand washers and vibrating screens in **CATIA V5**: parts/assemblies, 2D drawings, BOMs, dimensional tolerances/selective GD&T, shaft/bearing fits, material selection and multiple design revisions.
- Performed shaft/spring calculations and supported welded-structure fabrication; converted a vibrating screen from crankshaft/belt transmission to direct-excitation vibration motors to reduce transmission components and maintenance.

Working Student – Mechanical Design — Ario Fidar Maham Co., Tehran

11/2012–05/2013

- Recreated a 40x90 jaw crusher in **CATIA V5**, including individual components, assemblies and manufacturing drawings.

SELECTED DEVELOPMENT

Low-Cost Hand Prosthesis – Prototyping — CAD and 3D printing of a modular hand prosthesis; rebuilt a printer using Arduino/Marlin, hardware wiring, firmware/G-code changes and Slic3r process settings for ABS/PLA.

EDUCATION

M.Sc. Wind Energy Engineering, Hochschule Flensburg — current grade: 1.9

03/2023–11/2026

M.Sc. Mechanical Engineering, Hakim Sabzevari University — Control & Robotics

09/2014–01/2017

B.Sc. Mechanical Engineering, Islamic Azad University — Structural Dynamics

09/2008–06/2013

LANGUAGES

German: B2 (actively improving). English: C1 / IELTS 7.0. Persian: Native.